

Anisotropic Strength Analysis (SM Carbon, PAN)

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SM Carbon | $\Theta = 13.3^\circ$

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Input Data

- **Material Properties (SM Carbon, PAN)**
- $E_{11} = 145 \text{ GPa}$ $E_{22} = 10 \text{ GPa}$
 $G_{12} = 4.1 \text{ GPa}$ $\nu_{12} = 0.25$
- $R_{11}^T = 1520 \text{ MPa}$ $R_{11}^c = 1380 \text{ MPa}$
- $R_{22}^T = 41 \text{ MPa}$ $R_{22}^c = 170 \text{ MPa}$
- $T_{12} = 80 \text{ MPa}$
- **Orientation & Applied Stresses**
- $\Theta = 13.3^\circ$
- $\sigma_{xx} = 1528 \text{ MPa}$ $\sigma_{yy} = 268 \text{ MPa}$
 $\tau_{xy} = 625 \text{ MPa}$

Algorithm Overview

- **Stress Transformation** ($\sigma_{xx}, \sigma_{yy}, \tau_{xy} \rightarrow \sigma_{11}, \sigma_{22}, \tau_{12}$)
- **Hashin Failure Criterion**
 - Fiber mode (tension/compression)
 - Matrix mode (tension/compression)
 - $MS = \min(MS_{\text{fiber}}, MS_{\text{matrix}})$
- **Max Stress Criterion** (separate MS for 11,22,12)
- **Effective Strength** (R_{xx}, R_{yy}, T_{xy}) from Hashin MS
- **2D Failure Envelope** (σ_{xx} vs $\sigma_{yy}, \tau_{xy}=0$)
- **MS vs Θ** (0-90°, Hashin)
- **3D Failure Envelope** (σ_{xx} vs τ_{xy} for 5 σ_{yy} slices)

Stress Transformation & Hashin MS

- **Transformed Stresses ($\Theta=13.3^\circ$)**
 $\sigma_{11} = 1741.17 \text{ MPa}$
 $\sigma_{22} = 54.83 \text{ MPa}$
 $\tau_{12} = 276.76 \text{ MPa}$
- **Hashin Margin of Safety**
- Fiber MS = -0.1270
- Matrix MS = -0.7304 (matrix tension)
- **Total MS = -0.7304 → FAILED**

Mode	Margin of Safety
MS ₁₁ (fiber)	-0.1270
MS ₂₂ (transverse)	-0.2523
MS ₁₂ (shear)	-0.7109
Overall MS	-0.7109 → FAILED

Max Stress Criterion

Hashin gives slightly lower MS (-0.7304) because it couples transverse and shear stresses.

Effective Strength (Hashin)

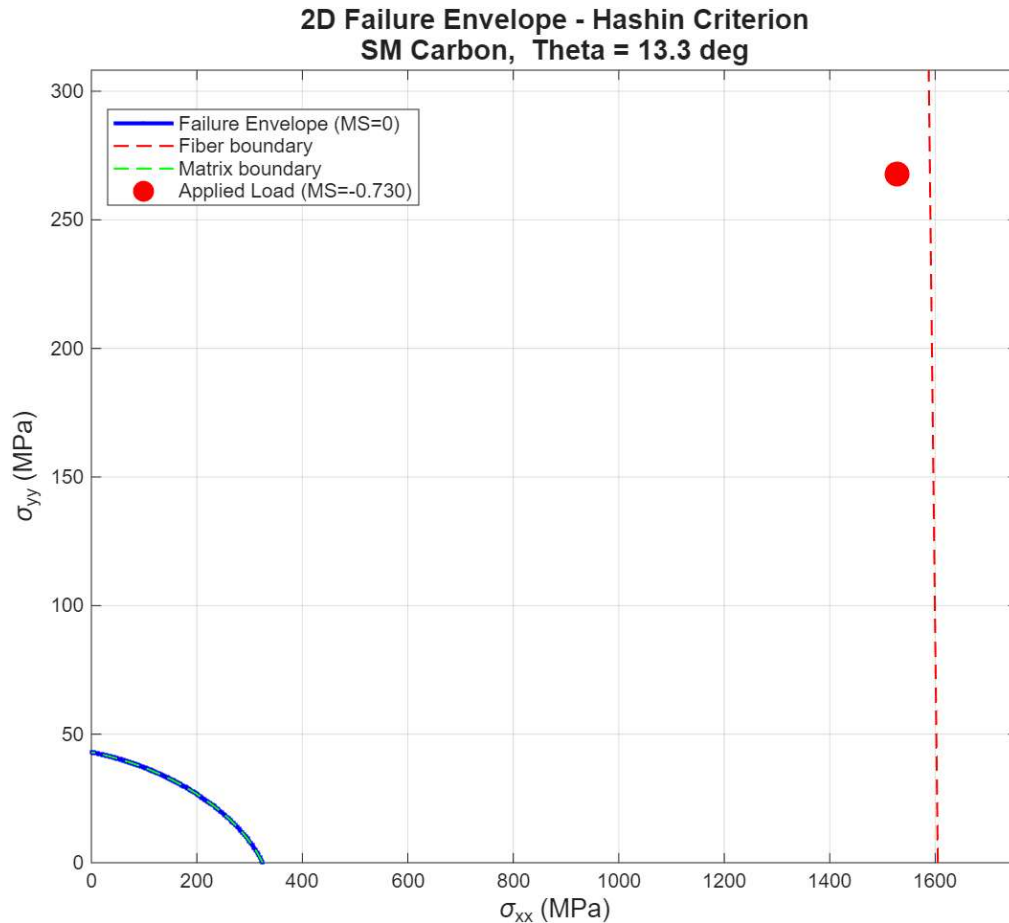
Using $MS = -0.7304 \rightarrow FS = 0.2696$

These are the stresses that would cause failure if the stress ratios (σ_{yy}/σ_{xx} and τ_{xy}/σ_{xx}) remain constant.

Strength	Value (MPa)
R_{xx} (σ_{xx} at failure)	411.97
R_{yy} (σ_{yy} at failure)	72.26
T_{xy} (τ_{xy} at failure)	168.51

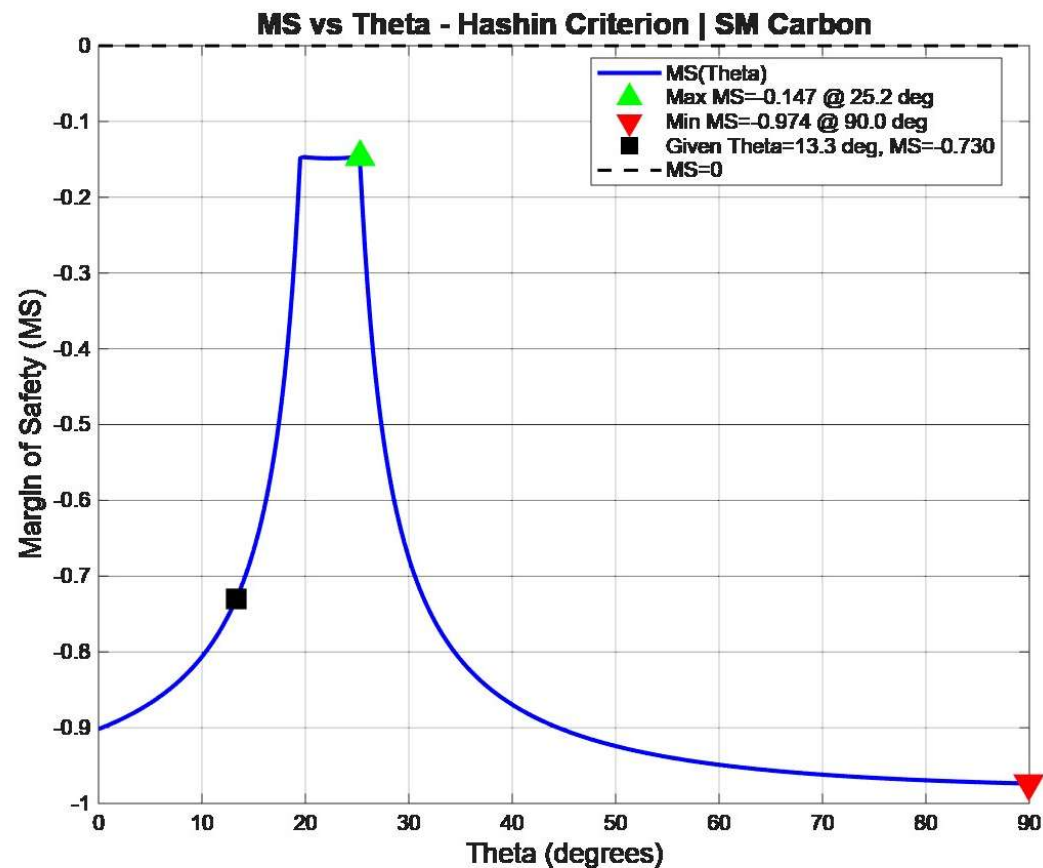
2D Failure Envelope (σ_{xx} vs σ_{yy} , $\tau_{xy}=0$)

- Blue solid line: Hashin failure envelope (MS=0)
- Red dashed: fiber boundary
- Green dashed: matrix boundary
- Red dot: applied load (1528,268) with MS = -0.730
- **Key observation:** Matrix failure dominates; the envelope reaches only $\sigma_{xx} \approx 324$ MPa at this orientation. The applied load lies far outside.



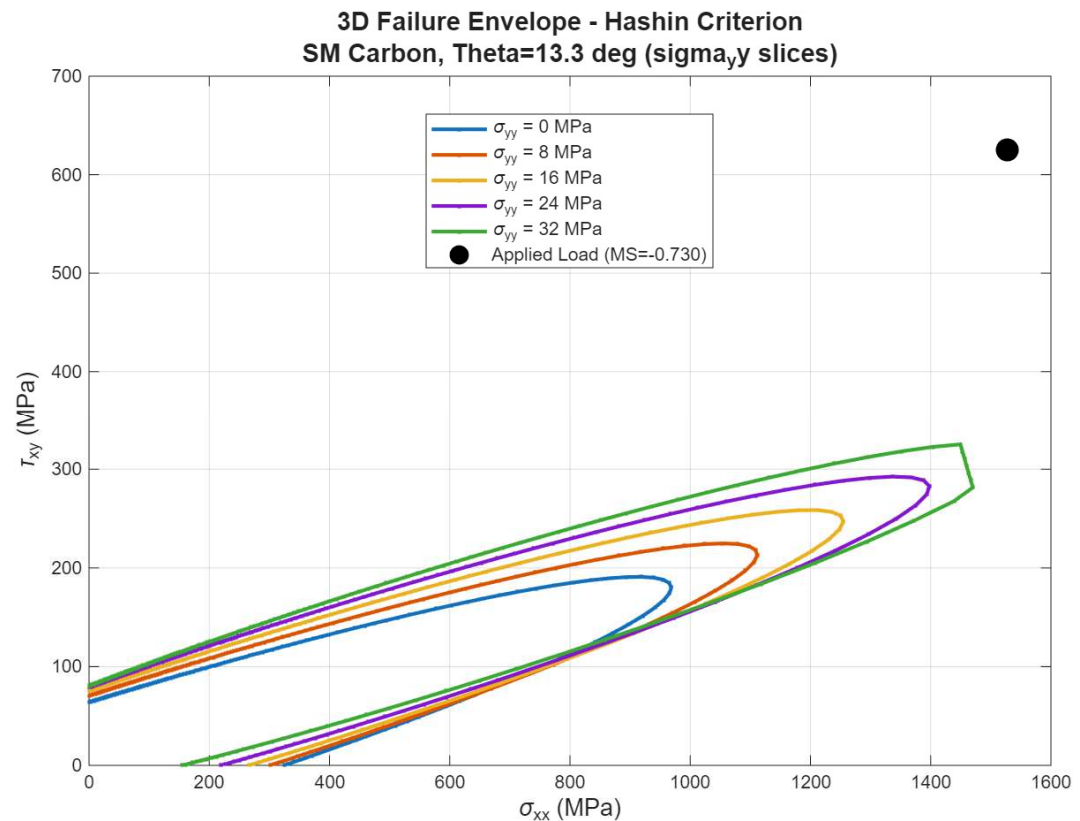
MS vs Orientation (Hashin)

- MS is **negative** for all Θ (load exceeds strength)
- Maximum (least negative) MS = -**0.147** at $\Theta \approx 25.2^\circ$
- Minimum MS = **-0.974** at $\Theta = 90^\circ$
- Given $\Theta = 13.3^\circ$, MS = -0.730
- Worst-case orientation is 90° (transverse load). Best is near 25° , but still unsafe.



3D Failure Envelope (σ_{xx} vs τ_{xy} , multiple σ_{yy})

- Five curves for $\sigma_{yy} = 0, 8, 16, 24, 32$ MPa
- Black dot: applied load (1528, 625) with MS = -0.730
- Envelope expands as σ_{yy} increases (higher transverse compression helps matrix)



Conclusions

- For SM Carbon at $\Theta = 13.3^\circ$, the applied load **fails** by both Hashin and Max Stress criteria ($MS \approx -0.73$).
- **Matrix tension + shear** is the critical failure mode.
- The **worst orientation** for this load is $\Theta = 90^\circ$ ($MS = -0.974$).
- The **least unsafe orientation** is $\Theta \approx 25^\circ$ ($MS = -0.147$), but still not safe.
- The 2D envelope ($\tau_{xy}=0$) is matrix-dominated, giving a small safe region.
- 3D envelope shows that increasing $\sigma_{\gamma\gamma}$ (compression) enlarges the safe region.

MATLAB

Code

- Stress transformation (xform)
- Hashin MS (hashin_MS)
- Max Stress MS (maxstress_MS)
- 2D envelope via radial search
- 3D envelope via analytical quadratic solution

```
%% =====  
% HMG - Anisotropic Strength Analysis  
% Krishiv A Patel | kap445  
% Material: SM Carbon | Theta = 13.3 deg  
% =====  
clear; clc; close all;  
  
%% INPUT DATA  
student_name = 'Krishiv A Patel';  
netid = 'kap445';  
  
E11 = 145e3;  
E22 = 18e3;  
G12 = 4.1e3;  
v12 = 0.25;  
R11T = 1520;  
R11C = 1380;  
R22T = 41;  
R22C = 170;  
T12 = 00;  
  
theta_deg = 13.3;  
theta_rad = deg2rad(theta_deg);  
sigma_xx_applied = 1520;  
sigma_yy_applied = 200;  
tau_xy_applied = 625;  
  
c_th = cos(theta_rad);  
s_th = sin(theta_rad);  
  
fprintf('=====\n');  
fprintf(' HMG | %s | SM Carbon | Theta=%1f deg\n', ...  
        student_name, netid, theta_deg);  
fprintf('=====\n\n');  
  
%% STRESS TRANSFORMATION  
[s11, s22, t12] = xform(sigma_xx_applied, sigma_yy_applied, ...  
        tau_xy_applied, theta_rad);  
fprintf('--- Stress Transformation ---\n');  
fprintf(' s11 = %1f MPa\n', s11);  
fprintf(' s22 = %1f MPa\n', s22);  
fprintf(' t12 = %1f MPa\n', t12);  
  
%% HASHIN CRITERION  
[MS_H, MSF, MSm, mstr] = hashin_MS(s11, s22, t12, R11T, R11C, R22T, R22C, T12);  
fprintf('--- Hashin Criterion ---\n');  
fprintf(' MS_fiber = %1f\n', MSF);  
fprintf(' MS_matrix = %1f\n', MSm);  
fprintf(' MS_total = %1f --> %s\n', MS_H, ...  
        ternary(MS_H>0, 'SAFE', 'FAILED'));  
  
%% MAX STRESS CRITERION  
[MS_MS, MS11, MS22, MS12] = maxstress_MS(s11, s22, t12, ...  
        R11T, R11C, R22T, R22C, T12);  
fprintf('--- Max Stress Criterion ---\n');  
fprintf(' MS_11 = %1f\n', MS11);  
fprintf(' MS_22 = %1f\n', MS22);  
fprintf(' MS_12 = %1f\n', MS12);  
fprintf(' MS_total = %1f --> %s\n', MS_MS, ...  
        ternary(MS_MS>0, 'SAFE', 'FAILED'));  
  
%% EFFECTIVE STRENGTH  
FS = MS_H + 1;  
Rxx = FS * sigma_xx_applied;  
Ryy = FS * sigma_yy_applied;  
Txy = FS * tau_xy_applied;  
fprintf('--- Effective Strength (Hashin) ---\n');  
fprintf(' Rxx = %1f MPa\n', Rxx);  
fprintf(' Ryy = %1f MPa\n', Ryy);  
fprintf(' Txy = %1f MPa\n', Txy);  
  
%% =====  
% 2D FAILURE ENVELOPE (sxx vs syy, txy = 0)  
% =====  
  
fprintf('Computing 2D Failure Envelope...\n');  
  
n_pts = 500;  
psi = linspace(0, pi/2, n_pts);  
ff_xx = zeros(1, n_pts); ff_yy = zeros(1, n_pts);  
fm_xx = zeros(1, n_pts); fm_yy = zeros(1, n_pts);  
env_xx = zeros(1, n_pts); env_yy = zeros(1, n_pts);  
  
for l = 1:n_pts  
    kxx = cos(psi(l));  
    kyy = sin(psi(l));  
  
    s11u = c_th^2*kxx + s_th^2*kyy;  
    s22u = s_th^2*kxx + c_th^2*kyy;  
    t12u = -c_th*s_th*kxx + c_th*s_th*kyy;  
  
    % Fiber FS  
end  
  
% 3D FAILURE ENVELOPE (optional, unchanged)  
% =====  
fprintf('Computing 3D Failure Envelope...\n');  
syy_slices = [0, 0, 16, 24, 32];  
n_slices = length(syy_slices);  
n_pts3 = 300;  
phi = linspace(1e-4, pi/2-1e-4, n_pts3);  
colors3 = lines(n_slices);  
  
figure('Name', '3D Failure Envelope', 'Position', [50 680 900 620]);  
hold on;  
for j = 1:n_slices  
    syy_f = syy_slices(j);  
    ex = nan(1, n_pts3); et = nan(1, n_pts3);  
    for l = 1:n_pts3  
        kxx = cos(phi(l)); kty = sin(phi(l));  
        A11 = c_th^2*kxx + 2*c_th*s_th*kty; B11 = s_th^2*syy_f;  
        A22 = s_th^2*kxx - 2*c_th*s_th*kty; B22 = c_th^2*syy_f;  
        A12 = -c_th*s_th*kxx + c_th^2 - s_th^2*kty; B12 = c_th*s_th*syy_f;  
        % Fiber mode (tension only)  
        if abs(A11) > 1e-12  
            sc_f = (R11T - B11)/A11;  
            if sc_f <= 0; sc_f = inf; end  
        else; sc_f = inf; end  
        % Matrix mode  
        if B22 >= 0; R22 = R22T; else; R22 = R22C; end  
        qa = (A22/R22)*2 + (A12/T12)*2;  
        qb = 2*(A22/R22/R22*2 + A12*B12/T12*2);  
        qc = (B22/R22)*2 + (B12/T12)*2 - 1;  
        sc_m = inf;  
        if qa > 1e-20  
            disc = qb^2 - 4*qa*qc;  
            if disc >= 0  
                r1 = (-qb + sqrt(disc))/(2*qa);  
                r2 = (-qb - sqrt(disc))/(2*qa);  
                pos_roots = [r1, r2]; pos_roots = pos_roots(pos_roots > 1e-6);  
                if isempty(pos_roots); sc_m = min(pos_roots); end  
            end  
        end  
        sc_env = min(sc_f, sc_m);  
        if isfinite(sc_env) && sc_env < 1e5  
            ex(l) = sc_env * kxx; et(l) = sc_env * kty;  
        end  
    end  
    plot(ex, et, '-', 'Color', colors3(j,:), 'LineWidth', 2, ...  
        'MarkerSize', 6, 'DisplayName', sprintf('\sigma_yy = %g MPa', syy_f));  
end  
plot(sigma_xx_applied, tau_xy_applied, 'ko', 'MarkerSize', 12, ...  
        'MarkerFaceColor', 'k', 'DisplayName', sprintf('Applied Load (MS=%1f)', MS_H));  
xlabel('\sigma_xx (MPa)', 'FontSize', 13);  
ylabel('\tau_xy (MPa)', 'FontSize', 13);  
title(sprintf('3D Failure Envelope - Hashin Criterion\nSM Carbon, Theta=%1f deg (sigma_yy slices)',  
        theta_deg), ...  
        'FontSize', 14, 'FontWeight', 'bold');  
legend('Location', 'best', 'FontSize', 10);  
grid on; box on; hold off;  
fprintf('\nAll done!\n');  
  
%% LOCAL FUNCTIONS  
function [s11, s22, t12] = xform(sxx, syy, txy, th)  
    c = cos(th); s = sin(th);  
    s11 = c^2*sxx + s^2*syy + 2*c*s*txy;  
    s22 = s^2*sxx + c^2*syy - 2*c*s*txy;  
    t12 = -c*s*sxx + c*s*syy + (c^2-s^2)*txy;  
  
end  
function [MS, MSF, MSm, mode_str] = hashin_MS(s11, s22, t12, R11T, R11C, R22T, R22C, T12)  
    if s11 >= 0; FS1 = R11T/s11; else; FS1 = R11C/abs(s11); end  
    if s22 >= 0; FS2 = R22T/s22; else; FS2 = R22C/abs(s22); end  
    FS12 = T12/abs(t12);  
    MS11 = FS1-1; MS22 = FS22-1; MS12 = FS12-1;  
    MS = min([FS11, FS22, FS12]) - 1;  
  
end  
function out = ternary(cond, a, b)  
    if cond; out = a; else; out = b; end  
end
```

```
if abs(s11u) > 1e-12  
    if s11u > 0  
        FSF = R11T / s11u;  
    else  
        FSF = R11C / abs(s11u);  
    end  
else  
    FSF = inf;  
end  
  
% Matrix FS  
if s22u >= 0; R22 = R22T; else; R22 = R22C; end  
denom_m = (s22u/R22)*2 + (t12u/T12)*2;  
if denom_m > 1e-20  
    FSm = 1/sqrt(denom_m);  
else  
    FSm = inf;  
end  
  
if isfinite(FSF)  
    ff_xx(l) = FSF*kxx; ff_yy(l) = FSF*kyy;  
end  
if isfinite(FSm)  
    fm_xx(l) = FSm*kxx; fm_yy(l) = FSm*kyy;  
end  
  
FS_env = min(FSF, FSm);  
if isfinite(FS_env)  
    env_xx(l) = FS_env*kxx;  
    env_yy(l) = FS_env*kyy;  
end  
end  
  
fprintf('--- 2D Envelope Debug ---\n');  
fprintf('R11T = %1f MPa\n', R11T);  
fprintf('Max envelope sxx = %1f MPa\n', max(env_xx(isfinite(env_xx))));  
fprintf('Max envelope syy = %1f MPa\n', max(env_yy(isfinite(env_yy))));  
  
figure('Name', '2D Failure Envelope', 'Position', [50 50 800 650]);  
plot(env_xx, env_yy, 'b-', 'LineWidth', 2, 'MarkerSize', 6, ...  
        'DisplayName', 'Failure Envelope (MS=0)');  
hold on;  
plot(ff_xx, ff_yy, 'r-', 'LineWidth', 1, 'DisplayName', 'Fiber boundary');  
plot(fm_xx, fm_yy, 'g-', 'LineWidth', 1, 'DisplayName', 'Matrix boundary');  
plot(sigma_xx_applied, sigma_yy_applied, 'ro', ...  
        'MarkerSize', 12, 'MarkerFaceColor', 'r', ...  
        'DisplayName', sprintf('Applied Load (MS=%1f)', MS_H));  
xlabel('\sigma_xx (MPa)', 'FontSize', 13);  
ylabel('\sigma_yy (MPa)', 'FontSize', 13);  
title(sprintf('2D Failure Envelope - Hashin Criterion\nSM Carbon, Theta = %1f deg',  
        theta_deg), 'FontSize', 14, 'FontWeight', 'bold');  
legend('Location', 'northeast', 'FontSize', 10);  
grid on; box on;  
  
% CRITICAL: Include both envelope and applied load in axis limits  
max_x = max([max(env_xx(isfinite(env_xx))), sigma_xx_applied]);  
max_y = max([max(env_yy(isfinite(env_yy))), sigma_yy_applied]);  
xlin([0, max_x * 1.15]);  
ylin([0, max_y * 1.15]);  
hold off;  
  
% =====  
% MS vs THETA  
% =====  
  
fprintf('Computing MS vs Theta...\n');  
theta_vec = linspace(0, 90, 361);  
MS_vec = zeros(size(theta_vec));  
for l = 1:length(theta_vec)  
    tr = deg2rad(theta_vec(l));  
    [s11, s22, t12] = xform(sigma_xx_applied, sigma_yy_applied, ...  
        tau_xy_applied, tr);  
    [MS_vec(l), ~, ~] = hashin_MS(s11, s22, t12, ...  
        R11T, R11C, R22T, R22C, T12);  
end  
  
[MS_max, lmax] = max(MS_vec);  
[MS_min, lmin] = min(MS_vec);  
fprintf(' Max MS = %1f at Theta = %1f deg\n', MS_max, theta_vec(lmax));  
fprintf(' Min MS = %1f at Theta = %1f deg\n', MS_min, theta_vec(lmin));  
  
figure('Name', 'MS vs Theta', 'Position', [870 50 750 580]);  
plot(theta_vec, MS_vec, 'b-', 'LineWidth', 2, 'DisplayName', 'MS(Theta)');  
hold on;  
plot(theta_vec, [MS_max, MS_min], 'g', 'MarkerSize', 12, ...  
        'MarkerFaceColor', 'g', ...  
        'DisplayName', sprintf('Max MS=%1f @ %1f deg', MS_max, theta_vec(lmax)));  
plot(theta_vec, [MS_min, ~], 'r', 'MarkerSize', 12, ...  
        'MarkerFaceColor', 'r', ...  
        'DisplayName', sprintf('Min MS=%1f @ %1f deg', MS_min, theta_vec(lmin)));  
plot(theta_deg, MS_H, 'ks', 'MarkerSize', 12, 'MarkerFaceColor', 'k', ...  
        'DisplayName', sprintf('Given Theta=%1f deg, MS=%1f', theta_deg, MS_H));  
yline(0, 'k', '-', 'LineWidth', 1.5, 'DisplayName', 'MS=0');  
xlabel('Theta (degrees)', 'FontSize', 13);  
ylabel('Margin of Safety (MS)', 'FontSize', 13);  
title('MS vs Theta - Hashin Criterion | SM Carbon', ...
```